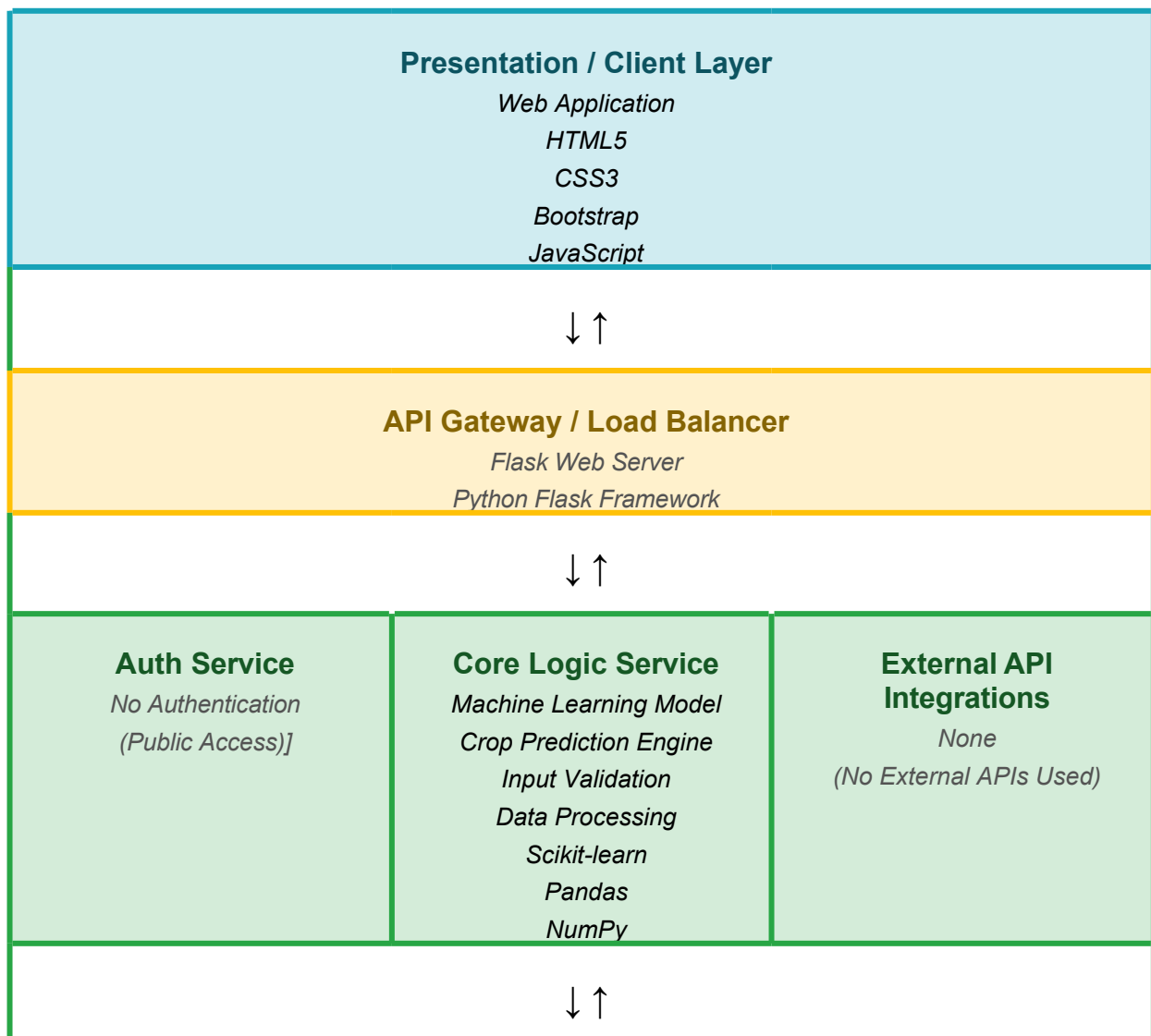


Solution Architecture

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Data / Storage Layer

Crop Recommendation Dataset (CSV)

Trained Model

crop_model.pkl

GitHub Repository

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides a user-friendly interface where users enter soil and environmental parameters and view crop recommendations.</i>	<i>HTML5, CSS3, Bootstrap, JavaScript</i>
<i>[API Gateway]</i>	<i>Receives HTTP requests from users and forwards them to the Flask application for processing.</i>	<i>Flask, Python</i>
<i>[Microservices]</i>	<i>Handles the core application logic, including validating user input, processing soil parameters, loading the trained machine learning model, and generating crop recommendations.</i>	<i>Python, Flask, Scikit-learn, Pandas, NumPy, Joblib</i>
<i>[Database]</i>	<i>Stores the crop recommendation dataset and the trained machine learning model used for prediction. The current version uses file-based storage instead of a database management system.</i>	<i>CSV Dataset (Crop Recommendation Dataset), crop_model.pkl, File System Storage</i>